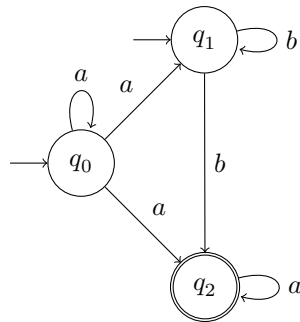


CS 2490 – Homework 4

Turn in by 2026/09/19 11:59PM

You **must** turn in your assignment as a single zip file entitled: *lastname_firstname_hw4.zip*. It will contain a pdf file *lastname_firstname_hw4.pdf* with your written answers (for which you can use $\text{\LaTeX}$, etc. or you can write your solutions by hand and scan them), and a JSON file for each relevant exercise named *lastname_firstname_hw4_exX.json* (where X is the exercise number).

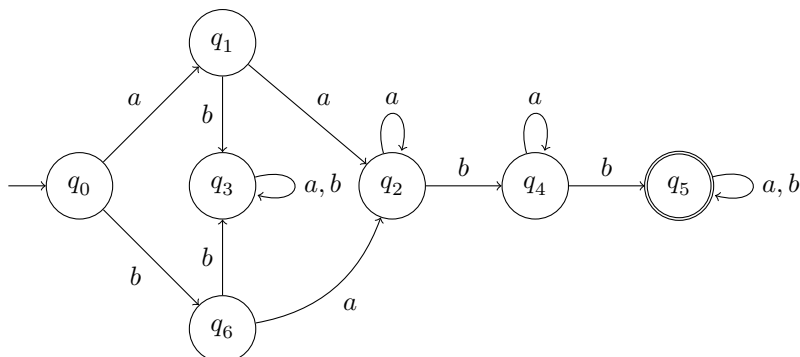
Exercise 1. Here is a NFA over the alphabet $\Sigma = \{a, b\}$:



Draw its determinization (the subset construction that makes it into a DFA). Use the DFA tool online to test your answer and export the JSON file that you will include in your submission.

Exercise 2. Call $\mathcal{A}$ the DFA you drew in the previous exercise. Is there any DFA with strictly fewer states than $\mathcal{A}$ that recognizes the same language? If yes, draw it. If no, explain why.

Exercise 3. Draw the minimization of the following DFA. (Include the table/half-table of distinguishability in your answer.)



Use the DFA tool online to test your answer and export the JSON file that you will include in your submission.

Exercise 4. Draw a NFA over the alphabet $\Sigma = \{a, b\}$ that recognizes the language of all words that contain the substring *bbab* anywhere.

Use the NFA tool online to test your answer and export the JSON file that you will include in your submission.

Exercise 5. Draw the determinization of the NFA you drew in the previous exercise.¹

Use the DFA tool online to test your answer and export the JSON file that you will include in your submission.

¹ Consistency will be taken into account. Thus, even if the previous exercise is not correct, you still get all credit for this exercise if you correctly determinize your NFA.

Exercise 6. Draw the minimization of the DFA you drew in the previous exercise.² Include in your answer the table/half-table of distinguishability.

Use the DFA tool online to test your answer and export the JSON file that you will include in your submission.

² Consistency will be taken into account. Thus, even if the previous exercise is not correct, you still get all credit for this exercise if you correctly determinize your NFA.

Exercise 7. Let Σ be any alphabet. We recall that the *reversal* of a word w , denoted by w^R , is the word obtained by reversing the order of the letters in w . That is, if $w = a_1 a_2 \cdots a_n$ with each $a_i \in \Sigma$, then $w^R = a_n a_{n-1} \cdots a_1$.

Show that if L is regular, then so is the language $L^R = \{w^R \mid w \in L\}$.